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Review

Does the US dollar confer an exorbitant privilege?



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ABSTRACT

This paper questions the asserted pecuniary benefits conferred by the dollar's international role. If these benefits are smaller than often claimed, then the difficult measurement of costs can be finessed. The original privilege notion characterised the dollar's systemic role in official foreign exchange reserves before 1973. Five subsequently developed notions refer to the broader, more evolutionary role of the dollar in the international monetary system. (1) As international debtor, the US borrows in its own currency – but others do too. (2) Dollar bills held offshore confer a benefit – but it is neither large nor exclusive. (3) The US Treasury may borrow more cheaply owing to official holdings – but it shares this advantage through an integrated global bond market. (4) The US earns higher yields on its external assets than it pays on its external liabilities – but this undoubted advantage arises from direct investment, far from the dollar's international role. (5) US banks may play on a home court – but they have in fact won a modest share of offshore dollar business.

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1. Introduction

“If I had an agreement with my tailor that whatever money I pay him he returns to me the very same day as a loan, I would have no objection at all to ordering more suits from him”. This simile of Jacques Rueff (1972, p. 78) captured the Gaullist view of the exorbitant privilege (EP). Not disciplined to settle debts in gold, the United States (US) exploited the 1960s gold-exchange standard to buy goods, services and whole companies with US dollar IOUs.

Many Chinese observers accept that the US dollar confers economic advantages on the US (Wang and Chin, 2013; Wang and Pauly, 2013). EP appeals because China makes net investment payments despite its international creditor status (Ma and McCauley, 2014; Wang, 2010) – while the US (the dual) receives such payments despite its debtor status. Certainly, Chinese officials hold lots of low-yielding US public bonds. Also, growing international use of the renminbi poses the question of benefits to China (Ba et al., 2010; Hai and Yao 2010; Genberg 2012).

EP's original meaning lost relevance in a world of floating fiat currencies, but EP has attracted new meanings. A second meaning holds that the US is privileged to finance its deficits in the dollar. The third through fifth meanings share the theme that the US dollar's role gives the US an easy ride in servicing its external liabilities. The sixth meaning holds that the dollar affords US financial firms a home-court advantage. In sum:

2. The US need not settle its dollar liabilities with some other asset.
3. The US can finance its current account deficits in its own currency.
4. The US runs up a zero-yielding debt to the rest of the world.
5. The US Treasury can borrow cheaply.
6. The US pays less on its external liabilities than it gets on its external assets.
7. US financial firms harvest “denomination rents” from global dollar use.

Yes, but. There is truth in each. But the privilege is small (Sections 4 and 7), not unique to the US (Sections 2–4), shared (Sections 4–6), or does not result from the dollar's international role (Section 6).

This article does not join a broader argument over whether the benefits of the dollar's international role exceed the costs (Aliber, 1964, 1966; Canzoneri et al., 2013; Cohen, 2012; Dobbs et al., 2009; Papaioannou and Portes, 2008; Salant, 1964). Cohen (2012) highlights the dollar's non-pecuniary benefits, including US prestige, akin to Hollywood's “soft power” (but see Williamson (2013).

Costs are not easy to measure. If the dollar's role has shrunk the US traded goods sector, productivity has grown more slowly (Belassa–Samuelson). Kindleberger (1973) identified leadership (for political scientists, “hegemonic”) costs: counter-cyclic demand, open markets for distress goods, international lender of last resort. Then Federal Reserve Bank of New York President Timothy Geithner alluded to the last during the Federal Open Market Committee (2008, p. 21) decision to extend swaps to Brazil, Korea, Mexico and Singapore: “the privilege of being the reserve currency of the world comes with some burdens”. This article does not try to measure costs but instead weighs pecuniary benefits in the above order.

2. No settlement

EP's first meaning is often widened: “the ability to buy foreign goods and companies using resources conjured out of thin air was the exorbitant privilege of which French finance Minister Valéry Giscard d'Estaing so vociferously complained” (Eichengreen, 2011, p. 40, *emphasis added*). However, the US current account surplus then meant that exports of goods and services paid for imports (Swoboda, 2012).

Americans debated whether the US gained policy flexibility (Aliber, 1964) or not (Salant, 1964) or both *seriatim* (Johnson, 1969). To Europeans, US gold not paying for US firms' European acquisitions gallingly added to a broader concern over US multinationals' organisational and technical prowess (Servan-Schrieber, 1967). Rueff stressed that European central banks were financing the competition.

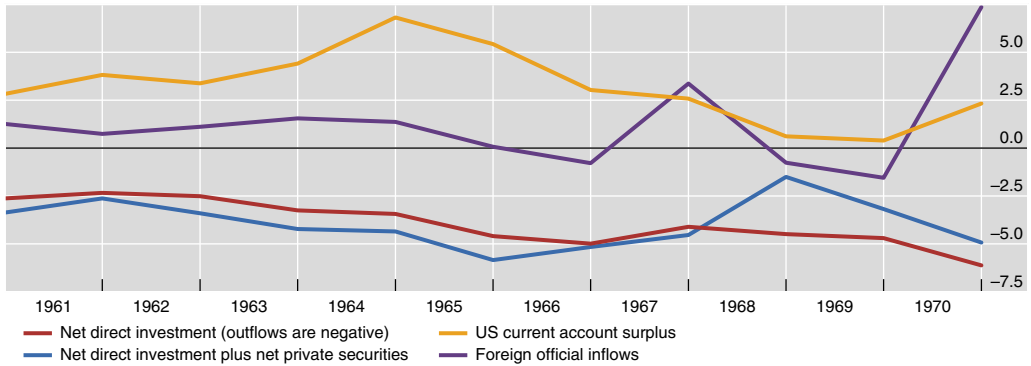


Fig. 1. US net long-term capital outflows, current account and official inflows, 1960–70 (in billions of US dollars). Source: US Bureau of Economic Analysis.

In fact, the US current account surplus (Fig. 1, yellow/dotted line) could have financed US multinational investment abroad (red/solid line) into 1965. And only in 1970 did foreign official flows provide regular support (purple/long dashed line).

Rueff's wily choice of simile, one of *The monetary sins of the West*, was confusing.¹ With cheques returned as loans, the customer buys more suits and runs a *trade* “deficit without tears”. Simile did not fit fact. More aptly: “if my stockbroker kept my cheques to settle his accounts, I would order more equities from him”.

President Nixon ended any pretence of gold convertibility on 15 August 1971 and rendered the original EP moot. Floating fiat currencies left only the likes of the Hong Kong dollar convertible into anything else. So yes, the US need not settle its dollar liabilities. But nowadays, no one does. However one reads the 1960s, no privilege now arises from unsettled dollars.

3. Own currency financing of current account deficits

Yes, the US finances its current account deficits in its own currency (Eichengreen, 2011, p. 5; Prasad, 2014, pp. 17, 114; Skidelsky, 2009, p. 182) – and claims on the rest of the world, too. The US ability to borrow dollars, to finance deficits and to invest in foreign currency assets around the world (Tille, 2003), is thought by many to give US policymakers a unique, even outrageous, opportunity to relieve itself of debt through the simple expedient of dollar depreciation. But it is not alone.

Though incorrect, it is plausible that the dollar's role confers a unique ability to finance accumulated deficits in its own currency. After all, the dollar's identified share of \$11.7 trillion in official reserves in September 2014 remains over three-fifths. Do these tailor loans explain the high share, around 95%,² of dollars in the \$28 trillion of US external liabilities in September 2014?

Not really: official liabilities are less important to the US as debtor than they were to the US as creditor. In fact, the US relied on foreign official creditors for 40% of its gross external liabilities in the

¹ The tailor is more apt than Rueff's marbles simile, where winners of US marbles return them at the end of play. This happened exceptionally during the first Gulf War. Meissner and Taylor (2006) twisted this simile towards insanity: “Losing our marbles in the new century?”

² This sums BEA direct investment and portfolio equity with the US Treasury reported dollar denominated US external debt www.treasury.gov/ticdata/Publish/debtb2014q3.html, taken as a share of the \$28 trillion, which excludes derivative positions (à la Lane and Shambaugh, 2010; Bénétrix et al., 2015). The US Treasury measures the share of dollars in currency-allocated external debt at 93%.

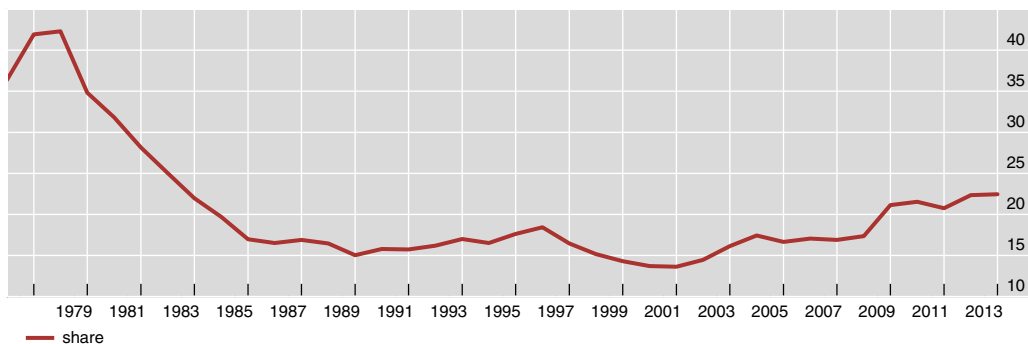


Fig. 2. Share of US liabilities to foreign officials in total US international liabilities (in per cent). Source: US Bureau of Economic Analysis.

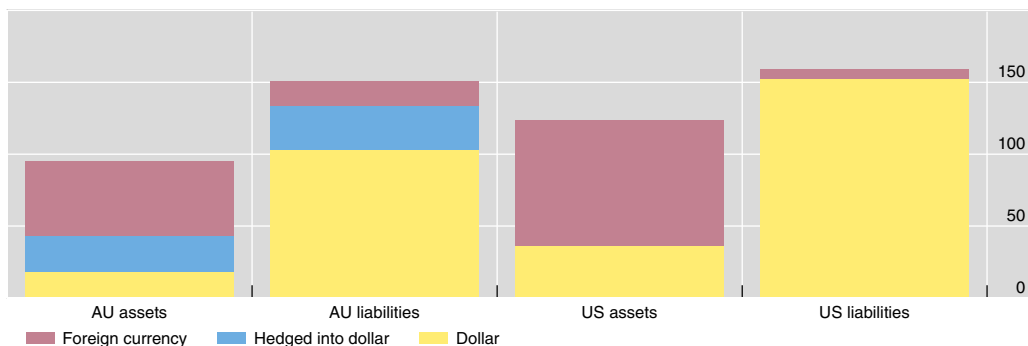


Fig. 3. Dollar and foreign currency external position in March 2013 (Australia) and September 2014 (US) (in percent of GDP). Sources: Rush et al. (2013, p. 51); Gohrband et al. (2014); US Treasury et al. (2014); author's calculations.

late 1970s before the onset of large and sustained current account deficits (Fig. 2). Since the US started to have net external liabilities, its dependence on official creditors has ranged around 15–20%.³

If the time series is not kind to the view that the dollar's reserve role allows the US to finance its deficits in the dollar, the cross-section rudely abuses the view. In relative terms, Australia has managed to finance a larger and longer string of current account deficits than the US (and to build up foreign currency assets as well) and it has also done so in domestic currency. In the process, Australia did not exploit the central role of its currency in global finance. The Australian dollar punches above Australia's weight (a tenth of the US economy) but it is only the fifth most traded with an identified share of currency reserves of less than 2%.

Despite its currency's bit role in global finance, Australia's external balance sheet's currency composition resembles that of the US (Fig. 3). Taking into account currency swaps, Australia had Australian dollar liabilities amounting to 130% of its GDP in 2013; the US had US dollar liabilities of 152% of its GDP. In both cases, dollar borrowing not only sufficed to finance the net external liability (60% of GDP for Australia and 35% for the US), but also to finance substantial foreign currency assets (in excess of half of GDP).⁴ One might argue that the demand for Australian dollar bonds is more concentrated (among

³ US liabilities understate the official role in dollar financing since central banks hold hundreds of billions of dollars in banks offshore (McCauley and Rigaudy, 2011).

⁴ Recent official reserve investment in the Australian dollar has not played a role in creating this currency composition of

Japanese households, for instance) and more fragile than demand for dollar bonds, but the flow of Australian bond issues was interrupted only briefly in late 2008.

Like the US, Australia's short domestic currency, long foreign currency position means that dollar depreciation reduces the country's international debt. The point is not just that advanced economies have the same structure to their external balance sheet (see [Ma and McCauley, 2014](#) for Germany). Instead a small country with a relatively larger net external liability position has such a structure. In view of this resemblance of the Australian and US external balance sheets, it is hard to see here a privilege that the US dollar's unique role confers on the US.

4. Seigniorage proper⁵

Yes, non-US residents hold hundreds of billions of greenbacks and these holdings are a boon to US taxpayers. But it is small and moreover is a duopoly with the euro.

Official estimates of the scale of non-resident holdings vary. [Nguyen \(2014\)](#) puts it at \$492 billion at end-2013. The [US Treasury \(2006\)](#), following [Judson and Porter \(2004, p. 285\)](#), estimated that 60% of all circulating US banknotes are held abroad. This implies \$806 billion held abroad at end-2014.

These holdings confer an interest-free loan to the US. The more dollars held in safe deposit boxes, safes and mattresses abroad as hedges against inflation ([Hellerstein and Ryan, 2011](#)) or expropriation, the more US banknotes circulate, and the more US Treasury bonds the Fed holds. After paying its costs, it remits its net income to the US Treasury. So the larger the demand for hundred dollar bills offshore, the more interest-free loans extended to the US Treasury.

How big is this benefit? "Seigniorage, or earnings from the assets that back the currency, totaled \$29 billion in 2005, or nearly all of Federal Reserve net income. That year, \$21 billion in earnings, about 1 percent of federal receipts, was remitted to the Treasury" ([US Treasury, 2006](#)). If 60% of the currency was held abroad, foreign residents contributed \$18 billion (60% of \$29 billion) of that \$21 billion, or about 0.9% of federal receipts.⁶ Since federal receipts approximate a fifth of GDP, the seigniorage from dollars held abroad was about 0.2% of GDP. "Seigniorage...is very small relative to other sources of revenues" ([King, 1995](#)).

At bond yields of say 2%, the high estimate of \$806 billion held abroad would have saved the US Treasury \$16 billion in 2014. While not large in budget terms, it would pay for the aircraft carrier strike group anchored in Yokosuka, Japan.

An external perspective also shows modest benefits: dollar banknotes held abroad are a small share of US international liabilities. Either the BEA estimate or the higher estimate based on 60% of US currency in circulation as the estimate of foreign holdings shows the interest-free share of US international liabilities to be small, measured in single-digit percentages ([Fig. 4](#)).

Moreover, the dollar shares this privilege with the euro area – and increasingly with the renminbi. Of the €950 billion in banknotes in circulation at the end of 2014, the ECB estimates that 20–25%, or €190–€240, are held outside the euro area.⁷ Thus euro area fiscal receipts benefit from currency held abroad less than half as much as the US fiscal receipts benefit from dollars held abroad. Renminbi banknotes are increasingly used in neighbouring countries, from Russia and Mongolia to Vietnam.

Whether split two ways or three ways, global seigniorage proper will remain modest. A privilege to be sure, but not a macro-economically significant one.

Australia's external position. Earlier surveys by [Becker and Fabbro \(2006\)](#) and [D'Arcy et al. \(2009\)](#), conducted presumably before the Australian dollar became so popular a reserve currency, show the same results qualitatively.

⁵ [Aliber \(1964\)](#) terms cheap Treasury borrowing ([Section 3](#)) seigniorage but [Johnson \(1969\)](#) opted for a narrower usage used here. [Cohen \(1971\)](#) measures the gap between the yield on official sterling balances and the return on UK overseas investment ([Section 4](#)). [Garber \(2012\)](#) highlights wartime use of US cash.

⁶ Note that the dollar benefit has not risen as much over the years as the foreign holdings of US currency as interest rates have fallen.

⁷ [Deutsche Bundesbank \(2015, p. 91\)](#) reports that half or more (75% of €156 = €118) of the euro banknotes outside of the euro area was shipped from Germany. The European authorities have inherited some of the Bundesbank's caution towards currency held abroad, which can confound interpretation of monetary aggregates ([Hardt, 2003](#)). US authorities facilitate such holding ([Botta, 2003](#)) and have even pre-placed dollars in a Hong Kong bank.

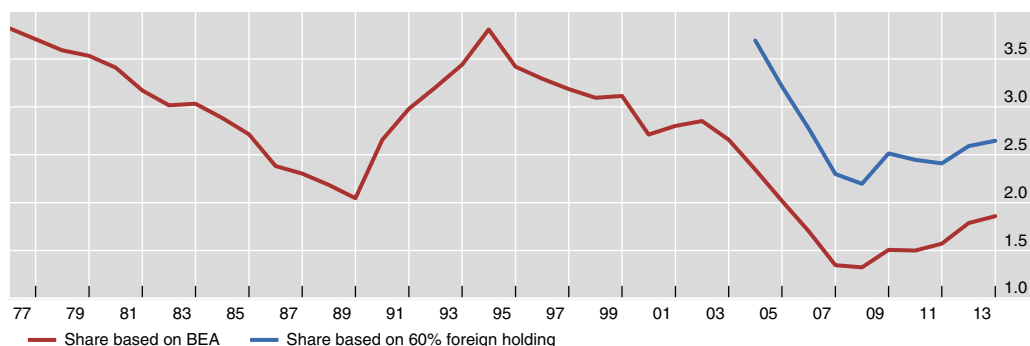


Fig. 4. Share of foreign holdings of US currency in total US international liabilities (in per cent). Sources: US Bureau of Economic Analysis, US Net International Investment Position and Federal Reserve Flow of Funds, table L108.

5. Cheap US Treasury borrowing?

The dollar's high share of official reserves may allow the US Treasury to borrow on the cheap (Bernanke et al., 2004; Dobbs et al., 2009; Eichengreen, 2011, p. 118; Jeanne, 2012, p. 1984; Rogoff and Tashiro, 2015; Warnock and Warnock, 2009). Yes, but this benefit is hard to measure (especially its persistence) and it is shared in three senses.⁸

First, not all reserves are held in dollars, so like seigniorage proper, this benefit is shared. While official reserves held in dollars are two to three times larger than those held in the euro, Truman (2012) reckons that this smaller euro reserve demand interacts with a smaller euro government debt stock to produce a cheapening of euro government debt of about half the number of basis points as the official demand for dollars. Calibrated at 30 basis points for US Treasury (see Dobbs et al., 2009), or 0.3% of US GDP, the saving on euro area government debt would amount to 16 basis points or 0.14% of GDP. Under this scenario, the UK government would also save 0.12% of GDP, and the Japanese, .04% of GDP.

Second, the benefit is shared by other issuers of US dollar bonds, including non-residents.⁹ Gerlach-Kristen et al. (2012) re-run Bernanke et al. (2004) and find that 10-year dollar interest-rate swaps responded one-to-one with Treasury bonds to Japanese daily intervention (ultimately largely invested in US Treasury bonds). Thus, any benefit diffuses to all US dollar bond issuers, including those issuers that are resident outside the US. International dollar bonds outstanding at end-September 2014 exceeded \$8 trillion, compared to a domestic US bond market of \$33 trillion.

And third, the benefit is shared to the extent that lower yields for US Treasury bonds lead to varying extents to lower yields for other governments' bonds in a globally integrated market. At the limit, if bonds in different currencies served as very close substitutes in private portfolios, it would not matter

⁸ Dobbs et al. (2009), citing Warnock and Warnock (2009), take this benefit to be 50–60 basis points. Genberg et al. (2005) pointed out an absurd implication of the Bernanke et al. (2004) estimate that each billion dollars of Japanese official purchase of dollars led to a 1 basis point lower 10-year US Treasury yield. Taken at face value, this estimate implies that the ¥35 trillion intervention lowered US bond yields by 300–400 basis points. The impact estimated on daily data must fade over time. Krishnamurthy and Vissing-Jorgensen (2012) calculate that the interest savings on Treasury securities from their liquidity and safety (to residents and non-residents) is of the same order of magnitude as seigniorage proper (on currency held by residents and non-residents).

⁹ If the effect were confined to the US Treasury bond market, one would expect to observe a wider spread between US bond yields and the generic private yields represented by interest-rate swaps. Given the absence of a euro-area wide sovereign issuer, the narrow spread between the German bunds and the euro interest rate swaps is generally interpreted as reflecting the relatively greater liquidity of euro swaps (Kreicher et al., 2014). The more comparable sterling bond market does not show a markedly narrower government-swap spread that would be expected if the official demand for US Treasuries reduced Treasury yields to the exclusion of other dollar bond yields. Dobbs et al. (2009) see other US borrowers as enjoying lower borrowing costs but overlook non-US issuers of dollar bonds.

Table 1

Estimates of spillovers of US bond yields to mature bond markets (basis points per 100 basis points on the US Treasury bond).

Bond market	Gerlach-Kristen et al. (2012): Japanese intervention, 2003–2004	Neely (2015): LSAP1 events		Bauer and Neely (2014): LSAP1 events	Rogers et al. (2014) intra-day data	Obstfeld (2015): long-term levels, monthly data 1989–2014):
	Government	Swap				
AU	53	60	67	37		74
CA	63	68	53	54		129
CH	61	53				88
DE	53	44	41	44	36	115
ES	53	44				111
FR	52	44				118
IT	52	44			16	158
JP	35	51	19	12	20	69
UK	65	49	46		48	137

AU = Australia; CA = Canada; CH = Switzerland; DE = Germany; ES = Spain; FR = France; IT = Italy; JP = Japan; UK = United Kingdom.

Notes: LSAP = first Federal Reserve large scale asset (ie bond) purchase. Gerlach-Kristen et al. (2012) estimates extended and updated by authors. Sources: Author's calculations based on cited work.

which bonds officials purchased. It would be wrong to imagine that this limiting case holds; but it would be even more wrong to imagine that a compression of Treasury yields stays in New York and does not touch yields in London, Frankfurt and even Tokyo.

The Federal Reserve's large-scale bond buying has inspired much analysis that suggests that buying of and announcements of buying of US Treasury bonds lead to substantial changes in yields on other sovereign bonds (Table 1; see also Chen et al., 2015). Neely (2015) finds that the 20–70% of the announcement effects of Federal Reserve bond buying diffused to mature bond markets. Bauer and Neely (2014) find that the effect is working through shared term premium (ie through channels other than correlated expectations of future short-term rates) in the larger German and Japanese markets. Rogers et al. (2014) confirm Neely's findings using high-frequency futures data. Again analysing the Japanese intervention of 2003–04, Gerlach-Kristen et al. (2012) find pass-through of a third to two-thirds of the effects on the Treasury market to other mature bond markets, including their interest rate swaps. Obstfeld (2015) estimates long-term level regressions that show that major government bond markets move in synch (even too much in synch) with the US Treasury market, with a median half-life of adjustment around one year.

To sum up, for the US Treasury to derive an exclusive privilege from official holdings of its bonds, the US dollar share of reserves would have to be close to one, other issuers of dollar bonds would have to not share in the Treasury's lower yields, and the global bond market would have to be unintegrated. But only 60% of reserves are held in the dollar, lower yield diffuse across the US dollar bond market, where non-resident issuers have a one-fifth share, and a third, a half or all of the bond yield compression in New York shows up in Frankfurt, Tokyo and London. A privilege so manifestly shared is hardly a privilege.

6. Rate of return advantage

The EP meaning that the US earns a higher rate of return on its external assets than it pays on its external liabilities not only enjoys widespread use (Eichengreen, 2011, p. 4; Gourinchas and Rey, 2007a, 2014; Habib, 2010; Jeanne, 2012, p. 1984; Meissner and Taylor, 2006; Rogoff and Tashiro, 2015), but also has entered a major textbook (Feenstra and Taylor, 2008). However, this “broader definition...confounds many different phenomena, not all of which can necessarily be thought of as ‘privilege’” (Rogoff and Tashiro, 2015). Yes, the US receives a higher rate of return on its external assets than it pays on its external liabilities, but nowadays it is hard to ascribe much of this return differential to the role of the US dollar.

Historically, the UK economy may have profited from sterling's role through oligopoly in the City of London and colony abroad. If India borrowed at 6% and deposited at 0% (de Cecco, 1984), then privileged was the UK.

Despres et al. (1966) saw such an asymmetry across maturity. US firm ITT profited from its French subsidiary, while US Treasury bills paid the Banque de France a pittance. Kindleberger (1965) argued that the US serves as a *global bank*, taking in liquid deposits and funding illiquid real assets. Aliber (1964) guesstimated that the US paid officials 3% and earned 10%. Europeans were not persuaded of a voluntary exchange, since the "bank" was appealing to military allies not to exchange deposits for gold (Kirschen, 1974; Oliva, 2009).

Gourinchas and Rey (2007a) revived Kindleberger's asymmetry, making out the US to be *global venture capitalist* with a leveraged and long equity position (Lane and Milesi-Ferretti, 2007). Unlike the venture capitalist, US debt rose in value in 2008 (Gourinchas et al., 2012), with the result that the US experienced massive mark-to-market losses on its external position. Gourinchas et al. (2010) shifted the simile to the US as *global insurer* that delivers mark-to-market gains to the rest of the world in the event of a crisis.

But with foreign acquisitions in the US the long US equity position is no longer so long. At end-2014, foreigners held \$12.8 trillion of direct and portfolio equity stakes in the US, while the US held \$13.8 trillion equity abroad (Gohrband et al., 2015). Dollar appreciation has narrowed the gap.

As a result, US investment income exceeds payments not so much because of the *mix across* US assets and liabilities, as because of return differences *within* asset classes. And in particular, the largest gap between earnings and payments within an asset class has long been in the direct investment accounts (Bosworth et al., 2007; Curcuru et al., 2013; Gohrband and Howell 2015; Higgins et al., 2005).

Some Europeans dismiss official data showing low returns of foreign firms in the US. Gourinchas and Rey (2007a, 2007b) assume that foreign firms' returns in the US match the returns of traded equity indices, in effect moving the puzzle elsewhere in the US external accounts (Curcuru et al., 2008). Gros (2006, p. 1) colourfully says:

it is unlikely that investors from all over the world would continue to pour hundreds of billions of dollars into foreign direct investment (FDI) in the US if they were really being constantly taken to the cleaners...

If foreign firms have used transfer prices to shift income out of the US, then the US rate of return advantage would be overstated (along with the US deficit on trade in goods and services but not the current account).¹⁰ Laster and McCauley (1994) found evidence that foreign firms in the US with more imports from their affiliates had lower returns, but Mataloni (2000) found no such evidence. Grubert (1997) found low rates of return among firms with less than majority ownership by foreign firms, which casts doubt on the income shifting since presumably the other shareholders would resist it. Landefeld et al. (1992), Grubert et al. (1993), Laster and McCauley (1994) and Mataloni (2000) and Tille (in Bosworth et al., 2007, pp. 202–204) all found evidence for returns rising with age.

Curcuru et al. (2013), drawing on Curcuru and Thomas (2015), ascribe only a third of the 5.6% gap between the returns on US direct investment abroad and direct investment in the United States to tax issues (1.8%). They draw attention to a definitional oddity, according to which US firms' foreign profits are reported gross of US taxes, while foreign firms' US profits are reported net of US taxes. US multinationals keep profits outside the US and try to persuade the US Congress of the benefits to the US economy of their repatriating these at a lower rates. Curcuru et al. assess that age accounts for 1½% of the difference: US firms' foreign affiliates are older than foreign firms' US affiliates. Aging improves returns because of start-up costs, higher initial reliance on debt and thin intellectual capital (Grubert et al., 1993). A third factor is risk, worth 0.8%: US firms invest in BBB-rated countries on average (Hung and Mascaro, 2004, but see Heath, 2007, p. 7).

¹⁰ With regard to US multinationals, Bernard et al. (2006) find that they price US exports to their affiliates abroad 43% below their arm's length prices and the gap is larger the lower the destination tax rate. Bosworth et al. (2007) find evidence for these firms' tax shifting to low-tax jurisdictions.

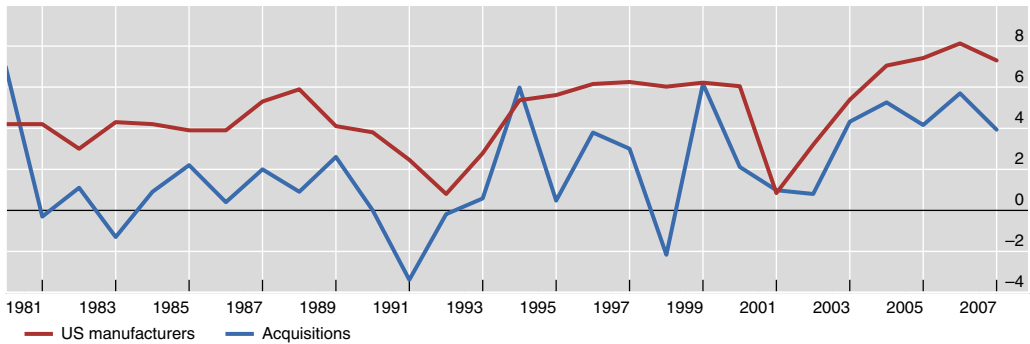


Fig. 5. Profitability of foreign acquisitions in manufacturing and of all US manufacturing (net income divided by sales in percent). Sources: US Bureau of Economic Analysis; Census; author's calculations.

Most analysis has neglected an important implication of expansion of foreign firms in the US by acquisition, as pointed out by Landefeld et al. (1992): adverse selection. This is consistent with work on corporate mergers: not only do acquiring firms' shareholders often lose but also many acquisitions are subsequently divested, suggesting pervasive regret. Cross-border acquisitions in the competitive US market for corporate control do not get the pick of the litter and are often divested.

Adverse selection is written all over accounting data on the acquisition targets (Fig. 5).¹¹ Non-US multinationals make half of their US acquisitions in manufacturing and their targets' returns are well below the average for US manufacturers. Mercedes acquired the long-troubled Chrysler, not Ford. The average return on sales of acquired manufacturers was less than half the US norm at 2% between 1980 and 2007 versus an average for all manufacturers of 4.8% (Fig. 5). Even if the low average return of foreign-acquired firms reflects a fat left tail of money losers rather than a difference in the mode (Laster and McCauley, 1994, Chart 6), foreign firms in the US cannot be thought of as buying the Standard and Poor's 500 index.

And the BEA finds high attrition rates of foreign direct investment in the US. In 1988, 3.8 million Americans worked for foreign firms. Over the nine years 1989 through 1997, foreign firms acquired US firms that employed 3 million workers. But over the same period, they sold back to US owners firms employing 2.3 million workers and closed down operations employing 0.9 million workers.¹² Overall employment at affiliates of foreign firms in the United States rose by just 1.4 million, thanks only to expansion of existing operations. Along with selecting adversely, paying premium prices and acquiring most heavily at valuation peaks, such attrition makes it plausible that foreign firms buying in the world's largest market for corporate control are taking themselves to the cleaners.

Contra Gros, the same firms need not repeatedly pour money into unsuccessful US acquisitions year after year. US firms may have made their mistakes abroad earlier (Lupo et al., 1978). In the US, European and Japanese firms made big acquisitions and now Brazilian, Chinese and Indian firms are taking their chances. Each wave of investors may end up with a rump of performing US operations, but the next wave's losses depresses the overall rate of return.

However one interprets the gap in direct investment returns, its connection to the US dollar's international role is remote. With the equity position of the US near balance, the days of exchanging

¹¹ Adverse selection has several aspects. Consistent with the Lichtenberg and Siegel (1992) finding of low productivity in manufacturing plants that are acquired, Doms and Jensen (1998) find that plants of foreign multinationals in the US are 4–8% less productive than the plants of US multinationals in the US. Mataloni (2000) finds that profitability is positively related to market share among US affiliates of foreign firms; low return on sales of acquired firms may indicate weak market positions.

¹² The BEA decomposition of sources of change in employment is limited to firms with over 500 employees, so it is not exhaustive. Unfortunately, it was discontinued before the all-time peak of foreign acquisitions of US firms in 1999–2000. The survey of US targets used in Fig. 5 was discontinued in 2008, but is to be revived.

Table 2

International bonds: book-runners and currency, 2009–2014 (issuance by nationals of neither the US nor euro area, in billions of US dollars).

Currency	Book-runner		Total
	US	Euro area	
Dollar	3,838	989	4,827
Euro	740	1,069	1,809
Total	4,578	2,058	6,636

Notes: The Chi-square statistic is 916, allowing the rejection of the null hypothesis of no relationship between currency of issue and nationality of book runner at the level of $p < .00001$. Sources: BIS; author's calculation.

Treasury bills for factories may live on for China (as it once held for Germany; [Ma and McCauley \(2014\)](#)) but not for the world.

7. Denomination rents

[Swoboda \(1968\)](#) proposed that financial firms headquartered in the key currency country derive extra profits from their home-court advantage, which he dubbed denomination rents. However, even at this early stage, he recognised the eurodollar market as a “means of distributing a part of the denomination rents...to the residents of other countries. That is, European [Japanese, Chinese...] banks, by issuing dollar liabilities, can bid away part of the gains from dollar denomination that previously accrued exclusively to American banks”. [Makin \(1976, p. 23\)](#) agreed, citing no barriers to entry. [Cooper \(2009\)](#) cites the success of London in both the dollar and the euro.

No longer on many economists' screens ([Eichengreen, 2011](#)), denomination rents loom large on those of political scientists. [Broz \(1997, p. 76\)](#) claims that US banks enjoy “a monopoly over the issue of [dollar] monetary liabilities”. [Frieden \(2003, p. 326\)](#) sees their advantage in “deeper financial markets and greater security” at home; [Cohen \(2012\)](#) in “privileged access to the resources of the issuing country's central bank”. [Helleiner \(2014, p. 76\)](#) places “powerful private financial firms on Wall Street...at the center of the selling of dollar securities to foreign investors...”

US banks' share of international dollar liabilities falls well short of a monopolist's at just 20.7% in September 2014 ([BIS Quarterly Review, March 2015](#), table A2). Highly competitive US money market mutual funds lowered the barriers to entry by funding non-US banks ([Baba et al., 2009](#)). Cohen's “privileged access” does not seem to count much.¹³

The advent of the euro tested Helleiner's hypothesis that the home court advantage counts in bond underwriting. The euro's introduction led to market share gains by US underwriters of bonds ([Santos and Tsatsaronis, 2003](#)). Regarding dollar bonds, [Galati and Tsatsaronis \(2003\)](#) find that US firms ran the books for about two-thirds of dollar-denominated international bonds in 1996–2000, but this includes US issuers' bonds. For only dollar bonds issued by nationals of the euro area, US firms ran the books for around 40% and euro area firms around 20%, suggesting that home-currency trumped home-country ties. Still, the remaining 40%, split among UK, Swiss and other firms with neither advantage, points to a contestable business.

How important has the home-currency advantage proved in 2009–2014? Issuers that are neither US nor euro area nationals sold \$6.6 trillion in dollar and euro-denominated bonds from 2009 to July 2014 using US or euro area book-runners ([Table 2](#), grand total). The bonds were 71% dollar-denominated and 69% of the book-runners were US, but did US firms disproportionately win dollar deals? This requires that US firms underwrote dollar bonds equal to more than half of the total (\$6.6 trillion $\times$ (71% $\times$ 69%) = \$6.6 trillion $\times$ 50% = \$3.3 trillion).

¹³ The US bank share is also low in lending dollars to non-bank borrowers outside the US ([McCauley et al., 2015, p. 196](#), footnote 14). Only since the financial crisis of 2007–2008 have risk averse official reserve managers placed more of their deposits with US-headquartered banks than with those of any other nationality ([McCauley and Rigaudy, 2011, Graph 3](#)).

The answer is yes: the home-currency advantage made a difference. In particular, US banks won \$0.5 trillion more in dollar bond business (\$3.8 trillion in the upper left of Table 2 less the expected \$3.3 trillion) than one would guess from the overall proportions. And, by construction, euro area firms won the same half trillion dollar equivalent more in euro bond business. In statistical terms, this is an overwhelming result. In economic terms, one could say that 8% of the business (\$0.5/\$6.6) went to US book-runners because of the dollar.

But this comparison overstates the home-court advantage because it neglects the British, Swiss and Japanese book-runners for the dollar and euro issues of the same borrowers. These bookrunners underwrote another \$6.1 trillion. On this showing, the home-court advantage (for \$0.5 trillion) made the difference for just 4% of the business.

In sum, there is something to the dollar providing a home-court advantage to US financial firms. But it provides a pretty subtle edge.

8. Conclusion

EP attracts new meanings over the years and the last several years have provided two new meanings. Goldberg (2013), drawing on Goldberg and Tille (2008), suggests that the fact that 95% of US trade is denominated in the dollar is a privilege. And Eichengreen (2015, p. 238) applies the term to the fact that foreigners flew to safety into US markets and the dollar in response to a US-centred crisis in 2008.

One recalls Cooper's distinction between effects of the size of the US and effects of the dollar's role. How much of the 95% share is owing to the size of the US economy (Ito and Chinn, 2014)? After all, two-thirds of euro area exports and one-half of its imports are euro-denominated and both shares are rising (European Central Bank, 2014, p. 9).¹⁴

Cooper's question is also pertinent to Eichengreen's new sense of EP as safe haven. The US Treasury bill market did indeed draw massive foreign investment in late 2008 (McCauley and McGuire, 2009). However, the liquidity of the US Treasury market that draws foreign investors in adverse circumstances reflects its size.¹⁵ In any case, these proposed meanings show that any assessment of EP confronts a moving target.

To conclude, the US as international debtor can borrow in its own currency and depreciate its way to a smaller debt. But this own-currency borrowing is not unique to the US and so cannot result from the dollar's unique role.

Hundred dollar bills held outside the US surely confer a benefit. But this is neither exclusive nor large.

The US Treasury may well borrow more cheaply as a result of the international role of the dollar. But this advantage is easy to overstate and crucially is shared through an integrated global bond market.

The US does enjoy an advantage in the rates of return on its international assets and liabilities. But this advantage lies mostly in foreign direct investment, quite remote from the US dollar's international role.

And US financial firms do have an advantage in operating in their home currency. But they do not have a corner on intermediation in the dollar, whether in banking or in bond markets.

¹⁴ See Genberg (2012) on the benefits to the home economy of invoicing in its currency. The high share of US trade invoiced in the dollar seems associated with more stickiness of dollar prices in the face of dollar exchange rate changes. As a result, currency changes may work on the US economy more through valuation effects on wealth than through relative price effects on trade (Gourinchas and Rey, 2007b). Does this make the US economy work better? As for the use of the dollar to invoice third-country trade, it supports dollar borrowing and dollar lending (including reserve holdings). The possible benefits of these to the US are analysed in Sections 1, 3, 4 and 5.

¹⁵ Moreover, in late 2008, the yen rose against the dollar, making it an even safer haven. For its part the dollar's rise in late 2008 resulted inter alia from European banks covering the losses on their private label US mortgages, a symptom of the "banking glut" of Shin (2012). Above, we have argued that the outsized role of non-US banks in dollar intermediation belies the notion that the dollar's role is a great boon to Wall Street.

A full consideration of the costs and benefits, pecuniary and non-pecuniary, of the dollar's role lies beyond this article's scope. What can be said is that it is easy to overstate how large and exclusive are the privileges, that is, pecuniary benefits, conferred by the dollar's role.

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